Comparing Air Pollution Trends Between Delhi, India and Dhaka, Bangladesh from 2003-2023

ENV 872L Final Project

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Introduction

Air pollution is one of the most pressing environmental and public health challenges of the 21st century, particularly across South Asia, where rapid urbanization, industrial growth, vehicular emissions, and agricultural waste burning have made several cities among the most polluted in the world (Ditta et al., 2026). Delhi, the capital of India, and Dhaka, the capital of Bangladesh, are two such megacities that consistently rank among the worst for air quality globally, with both cities experiencing severe seasonal pollution cycles driven by winter temperature inversions, stubble burning, brick kilns, and vehicle emissions along the shared Indo-Gangetic Plain (Ditta et al., 2026; Begum et al., 2018). Despite their geographic proximity and shared pollution sources, the two cities have followed divergent regulatory paths. India launched its National Clean Air Programme in 2019 targeting significant reductions in particulate matter (Garg et al., 2022), while Bangladesh was among the first South Asian nations to ban two-stroke engine vehicles in the early 2000s and transition to CNG-powered alternatives (State of Global Air, n.d.), raising important questions about whether these differing policy trajectories have translated into meaningfully different pollution trends over time.

While prior studies have individually examined air quality trends in Delhi (Garg et al., 2022; Singh et al., 2023) and Dhaka (Islam et al., 2021; Begum & Hopke, 2018), long-term comparative analyses spanning both cities remain limited. This project addresses that gap by using 20 years of monthly air quality data (2003 to 2023) to compare air quality trends in PM2.5, PM10, ozone (O3), and AQI across both cities, testing whether there is a significant seasonal component or decreasing trend, whether Delhi's pollution is significantly worse than Dhaka's, and whether the two cities are improving at statistically different rates. Thus, this study addresses the following hypotheses:

H_1: Air quality in Delhi and Dhaka exhibits a statistically significant seasonal trend over the 2003–2023 study period.

H_2: Air quality in Delhi and Dhaka exhibits a statistically significant long-term decreasing trend over the 2003–2023 study period.

H_3: Delhi's air quality is significantly worse than Dhaka's, as measured by mean concentrations of PM2.5, PM10, O3, and AQI over the 2003–2023 study period.

Data description

Dhaka Dataset

The Dhaka air quality data was sourced from the Bangladesh Air Quality Index (AQI) Dataset (2000–2025), published by Tapon Paul of Daffodil International University and retrieved from Mendeley Data (Paul, 2026). The full dataset contains 1,048,551 hourly records spanning 103 cities across Bangladesh across 13 columns: city_id, city_name, lat, lon, datetime, pm10, pm2_5, carbon_monoxide, carbon_dioxide, nitrogen_dioxide, sulphur_dioxide, ozone, and aqi. AQI values are calculated according to the US EPA standard.

Delhi Dataset

The Delhi air quality data was sourced from the Processed Delhi Air Quality and Weather Data dataset curated by Abhinav Sarkar and retrieved from Hugging Face (Sarkar, 2024). The full dataset contains approximately 2.9 million hourly observations from multiple monitoring locations across Delhi from March 2000 to November 2024, across 15 columns: city, event_timestamp, temperature, humidity, pressure, wind_speed, wind_direction, pm25, pm10, no2, so2, o3, co, aqi, and location_id. The dataset was pre-processed prior to publication, with missing AQI values removed and remaining gaps addressed through mean imputation and nearest neighbor interpolation.

Methods

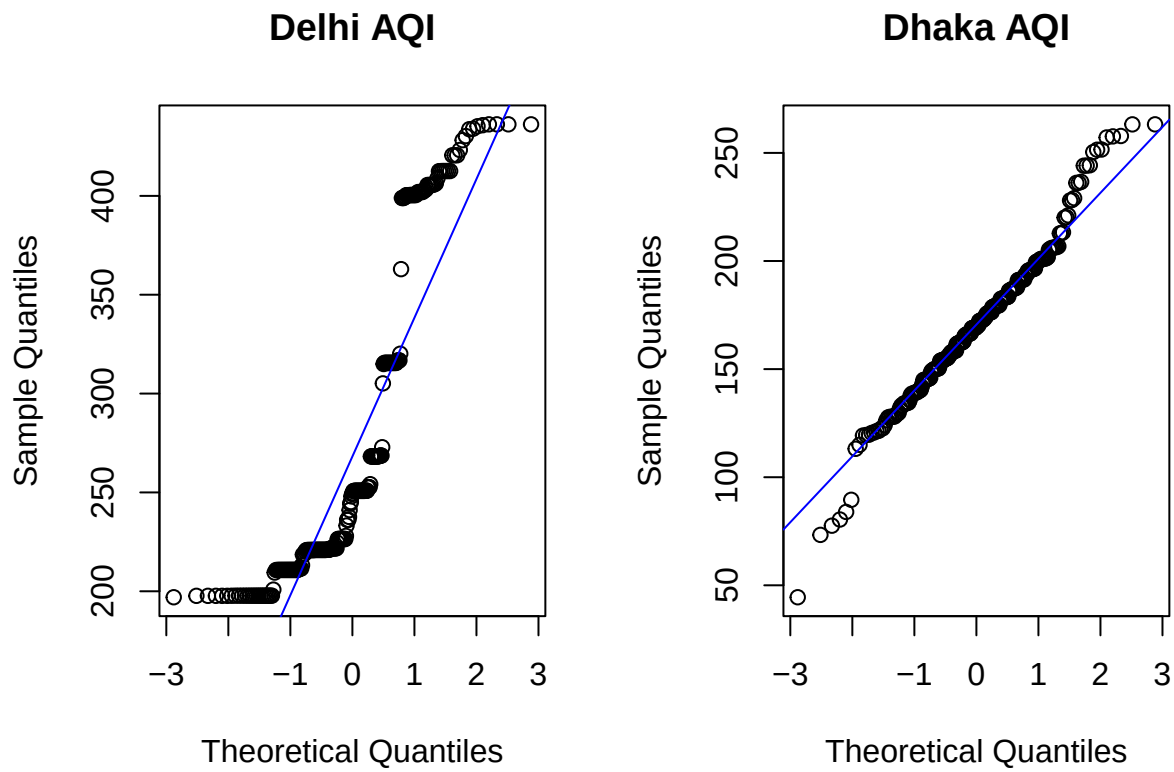
Initial Data Wrangling

Both datasets were cleaned and processed separately before being combined for analysis. For the Dhaka dataset, we filtered the to retain only observations from Dhaka city. For both datasets, the datetime columns were converted into a proper R datetime object using the ymd_hms function and only the columns needed for analysis were selected: Location, Month, Year, PM2.5, PM10, O3, and AQI. Both datasets were filtered to January 1, 2003 through December 31, 2023. We chose this window because the Dhaka dataset spans a longer date range from 2000 to 2025 and the Delhi dataset does not have complete monthly records for 2000 and 2024. To keep the data consistent and comparable across both cities, we focused on a clean 20-year trend. Hourly observations for both datasets were aggregated into monthly averages for all four pollutants (AQI, PM 2.5, PM 10, and O3) using na.rm = TRUE to handle any missing values. A Location variable was added to each dataset before merging and the two datasets were combined using rbind. A date column marking the first day of each month was created using make_date for use in visualizations and regression analyses. The final combined dataset contained 504 monthly observations over 8 columns (see below).

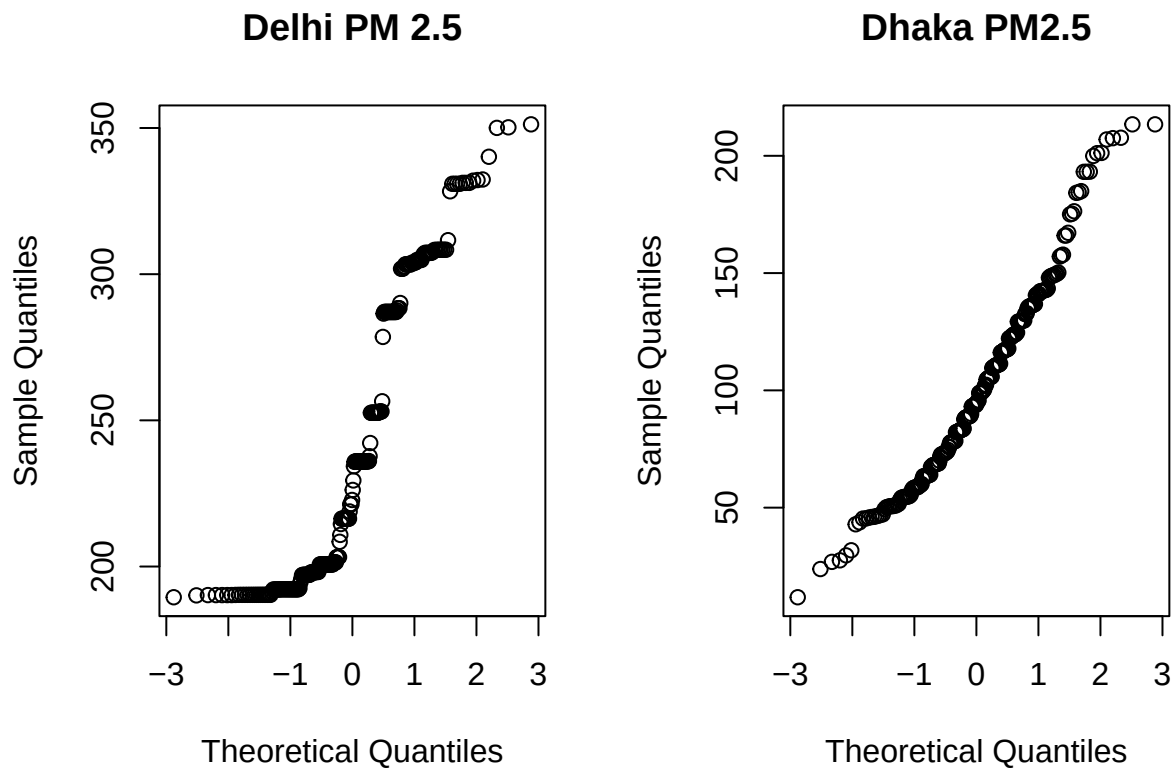
```
## 'data.frame':    504 obs. of  8 variables:
##  $ month      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ year       : int  2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 ...
##  $ PM2.5      : num  70 74.6 82.5 88.9 95.5 ...
##  $ PM10       : num  119 127 139 151 162 ...
##  $ O3         : num  42.4 42.8 43.2 42.5 42.6 ...
##  $ AQI        : num  151 156 162 166 170 ...
##  $ Location   : chr  "Dhaka" "Dhaka" "Dhaka" "Dhaka" ...
##  $ first.month: Date, format: "2003-01-01" "2004-01-01" ...
```

```
##       month        year       PM2.5        PM10          O3         AQI
##           0           0           0           0           0           0
##    Location first.month
##           0           0
```

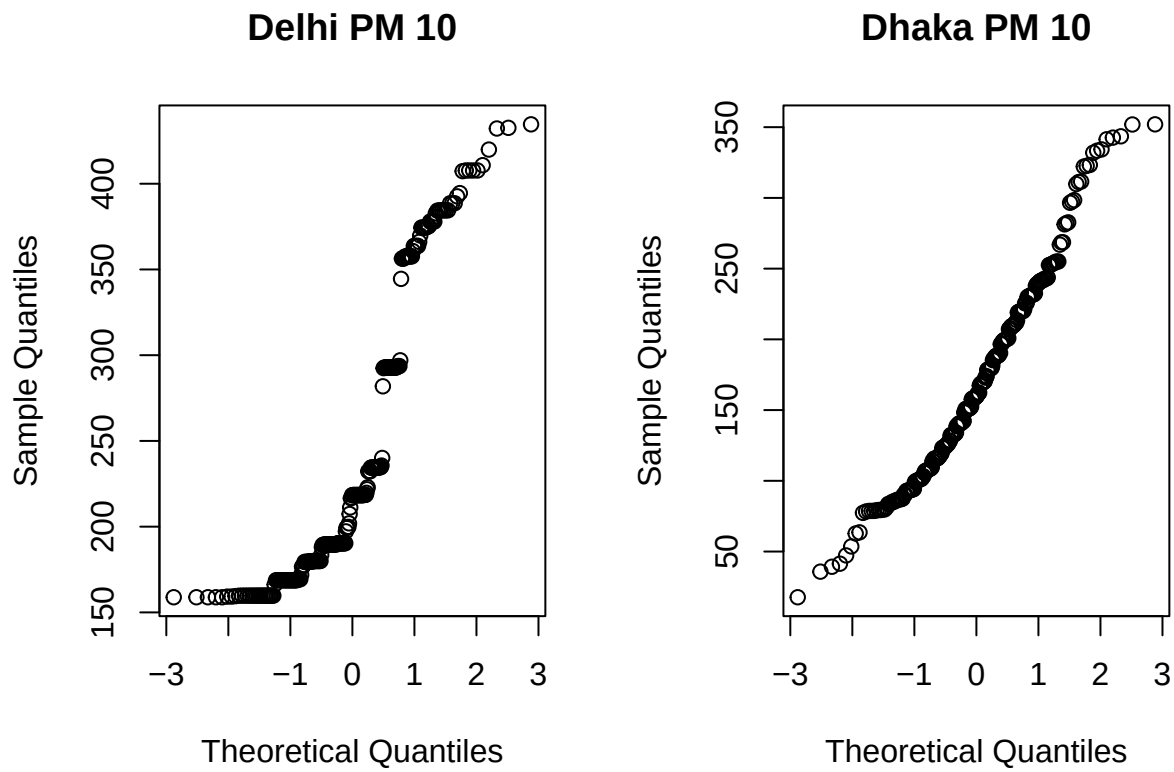
```r
par(mfrow = c(1, 2))
qqnorm(combined_dataset$AQI[combined_dataset$Location == "Delhi"],
       main = "Delhi AQI")
qqline(combined_dataset$AQI[combined_dataset$Location == "Delhi"],
       col = "blue")
qqnorm(combined_dataset$AQI[combined_dataset$Location == "Dhaka"],
       main = "Dhaka AQI")
qqline(combined_dataset$AQI[combined_dataset$Location == "Dhaka"],
       col = "blue")
```



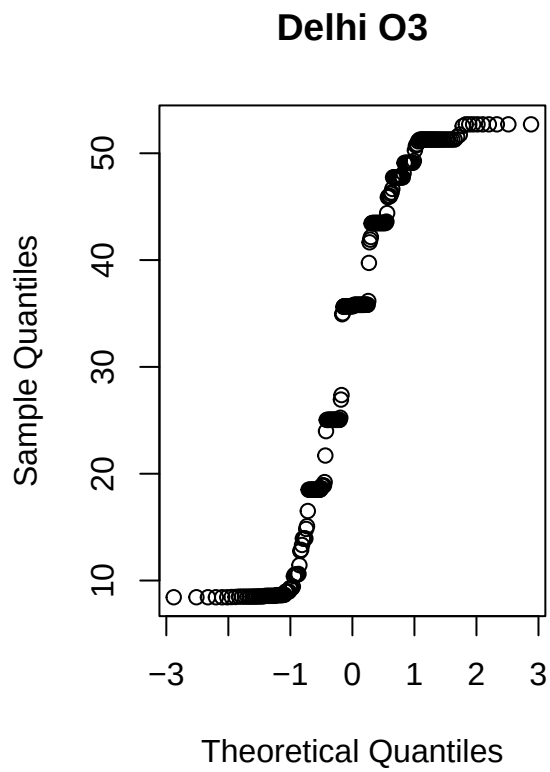

```
par(mfrow = c(1, 2))
qqnorm(combined_dataset$PM2.5[combined_dataset$Location == "Delhi"],
       main = "Delhi PM 2.5")
qqnorm(combined_dataset$PM2.5[combined_dataset$Location == "Dhaka"],
       main = "Dhaka PM2.5")
```

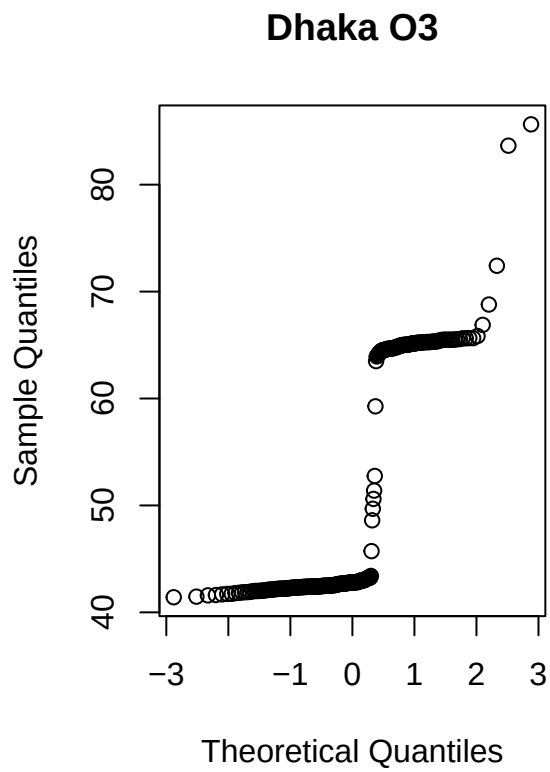



```r
par(mfrow = c(1, 2))
qqnorm(combined_dataset$PM10[combined_dataset$Location == "Delhi"],
       main = "Delhi PM 10")
qqnorm(combined_dataset$PM10[combined_dataset$Location == "Dhaka"],
       main = "Dhaka PM 10")
```




```
par(mfrow = c(1, 2))
qqnorm(combined_dataset$O3[combined_dataset$Location == "Delhi"],
       main = "Delhi O3")
qqnorm(combined_dataset$O3[combined_dataset$Location == "Dhaka"],
       main = "Dhaka O3")
```





Results

Discussion

Limitations

Conclusion

References

TIME SERIES ANALYSES

```
# Monthly AQI for Delhi
Delhi.AQI.monthly.ts <- ts(delhi_final_processed$AQI,
                           start = c(2003,1), frequency = 12)
Delhi.AQI.monthly.decomposed <- stl(Delhi.AQI.monthly.ts,
                                    s.window = "periodic")
plot(Delhi.AQI.monthly.decomposed)
```

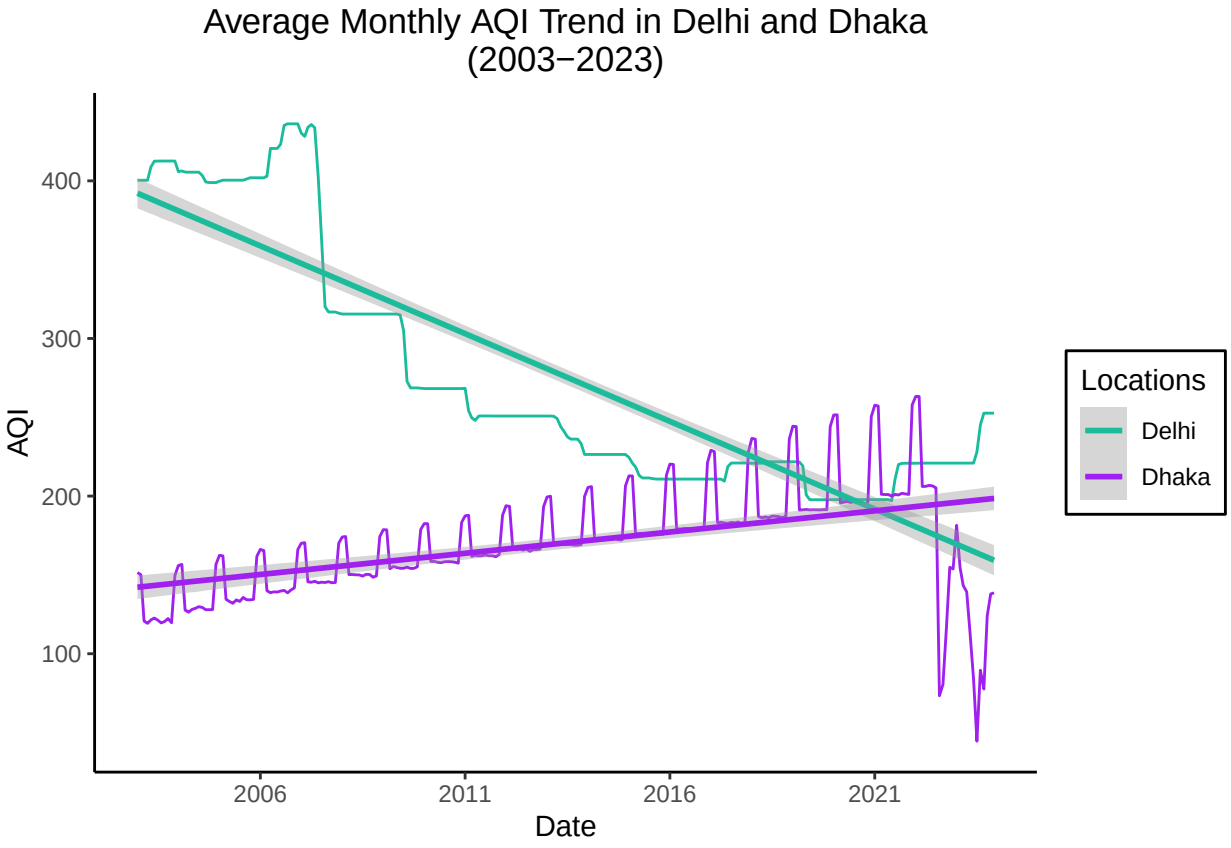


Figure 1: Monthly AQI Trend from 2003-2023

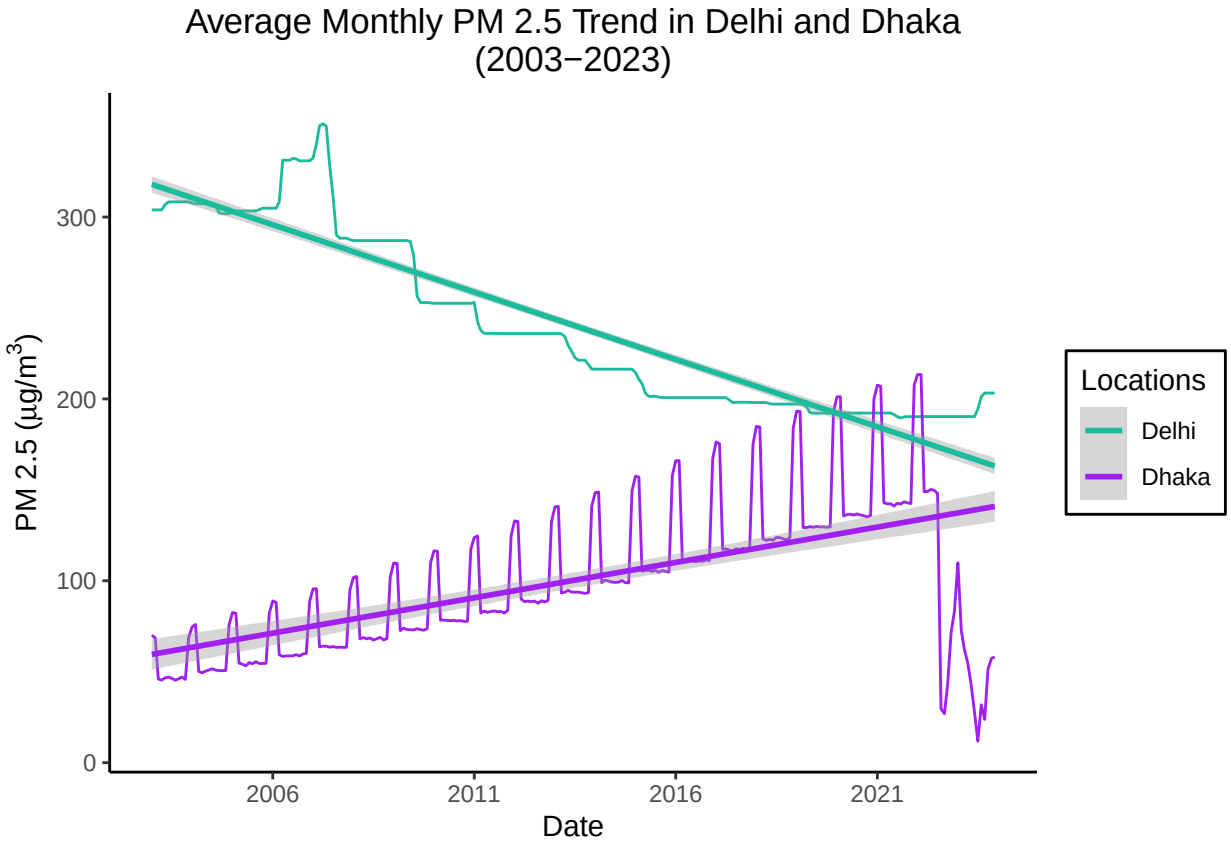


Figure 2: Monthly PM 2.5 Trend from 2003-2023

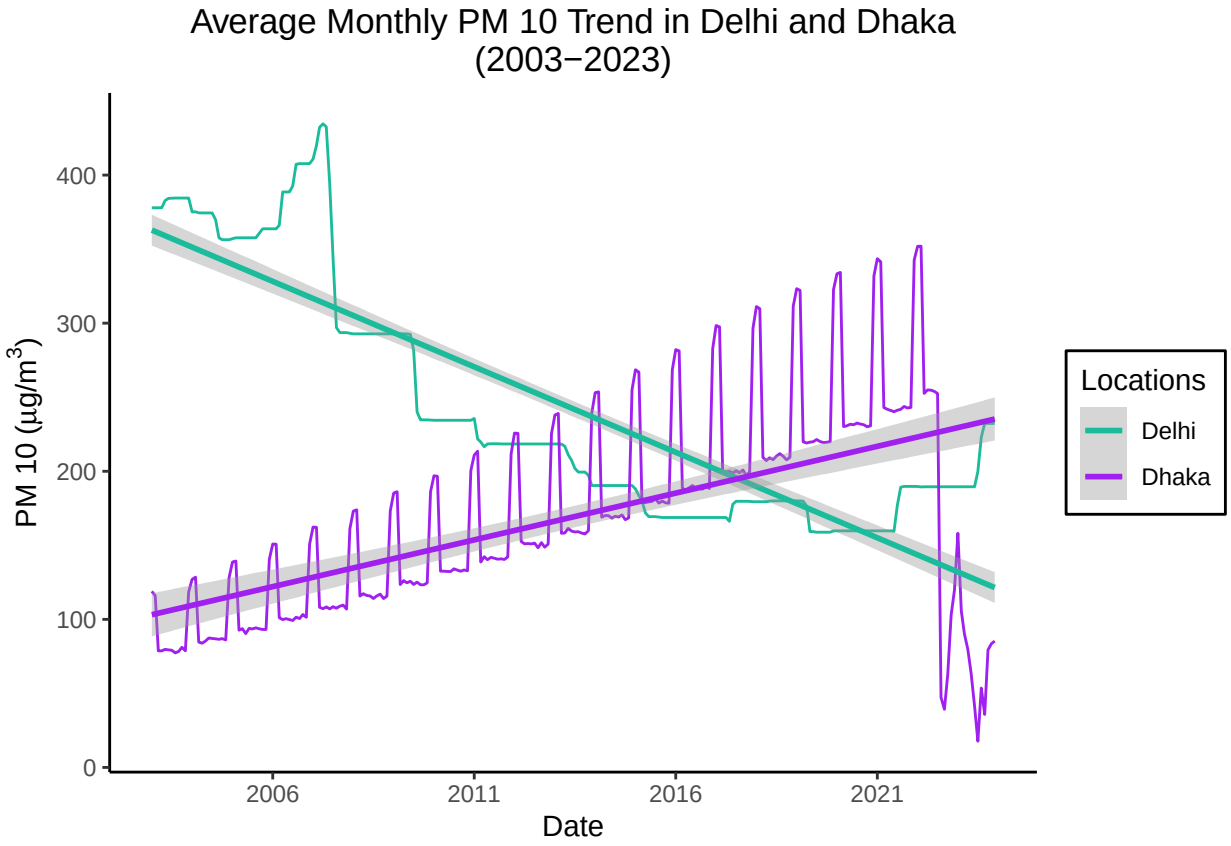


Figure 3: Monthly PM 10 Trend from 2003-2023

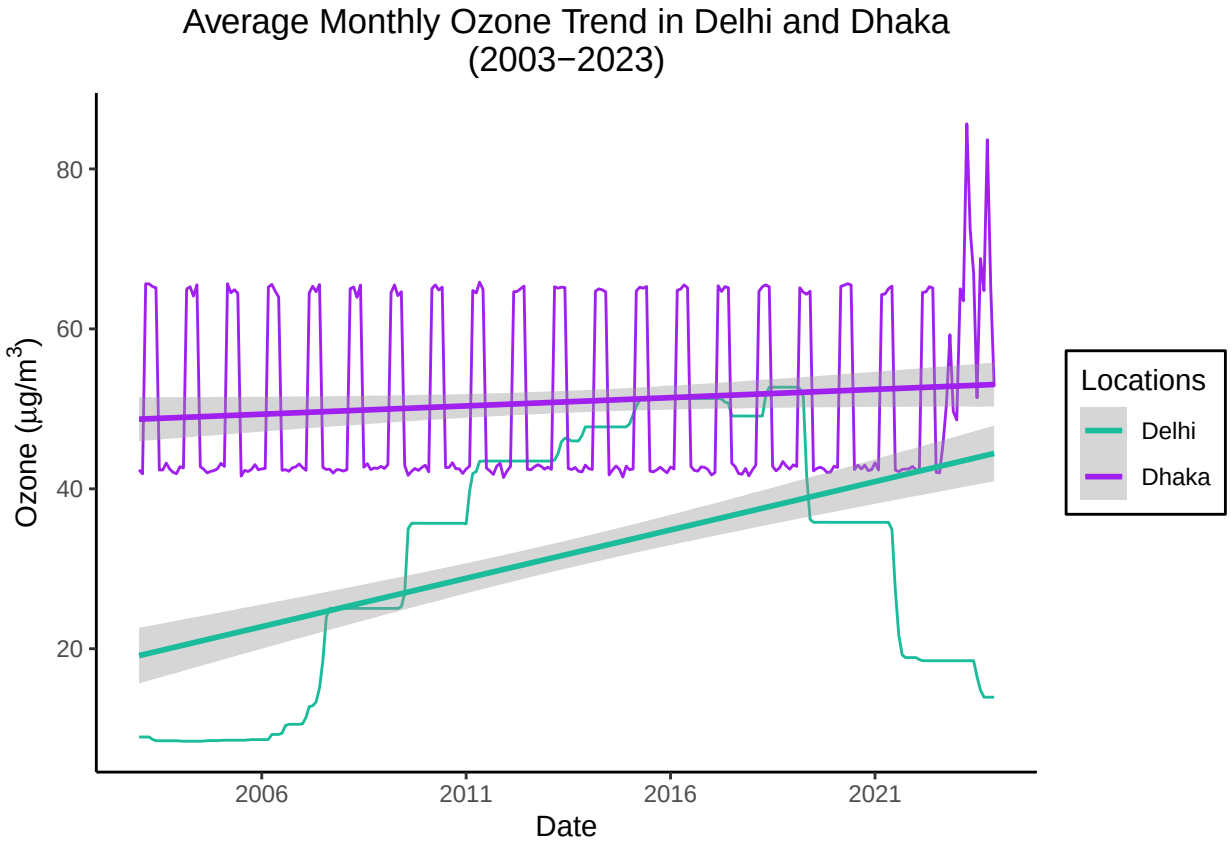
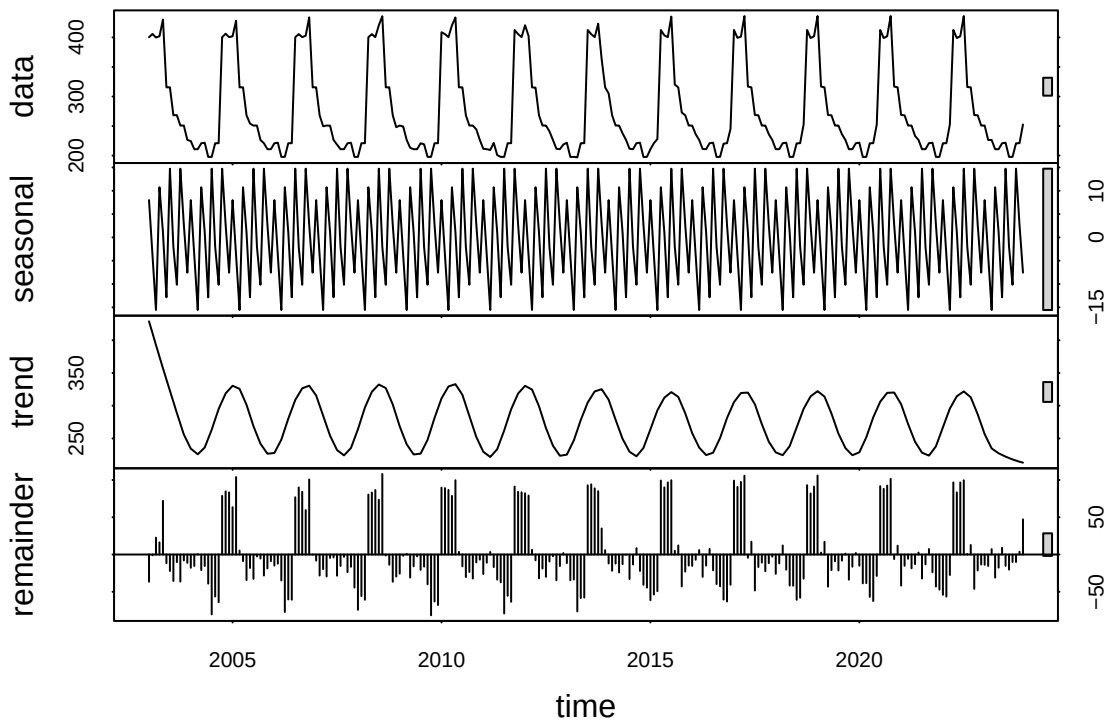
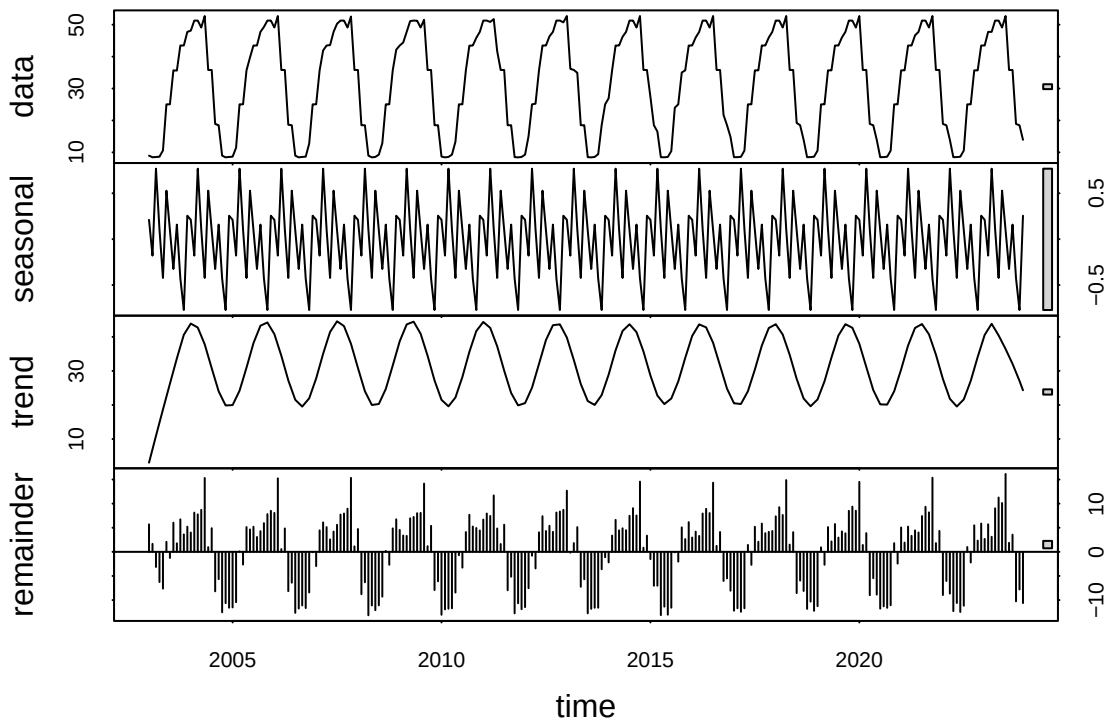


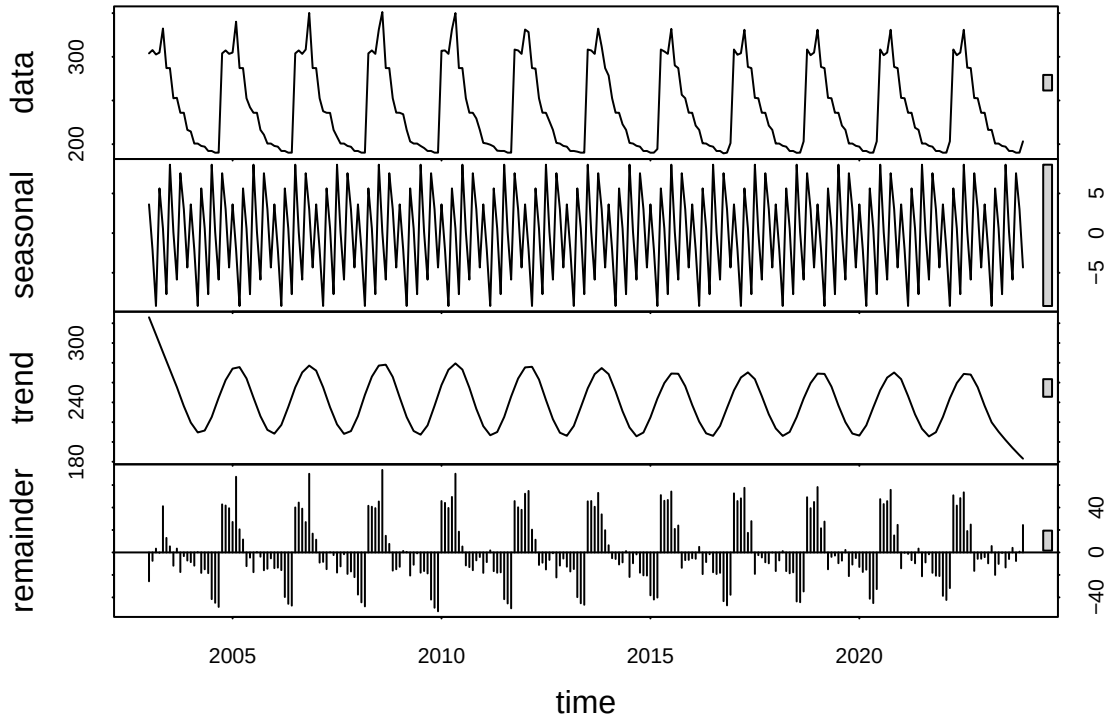
Figure 4: Monthly Ozone Trend from 2003-2023



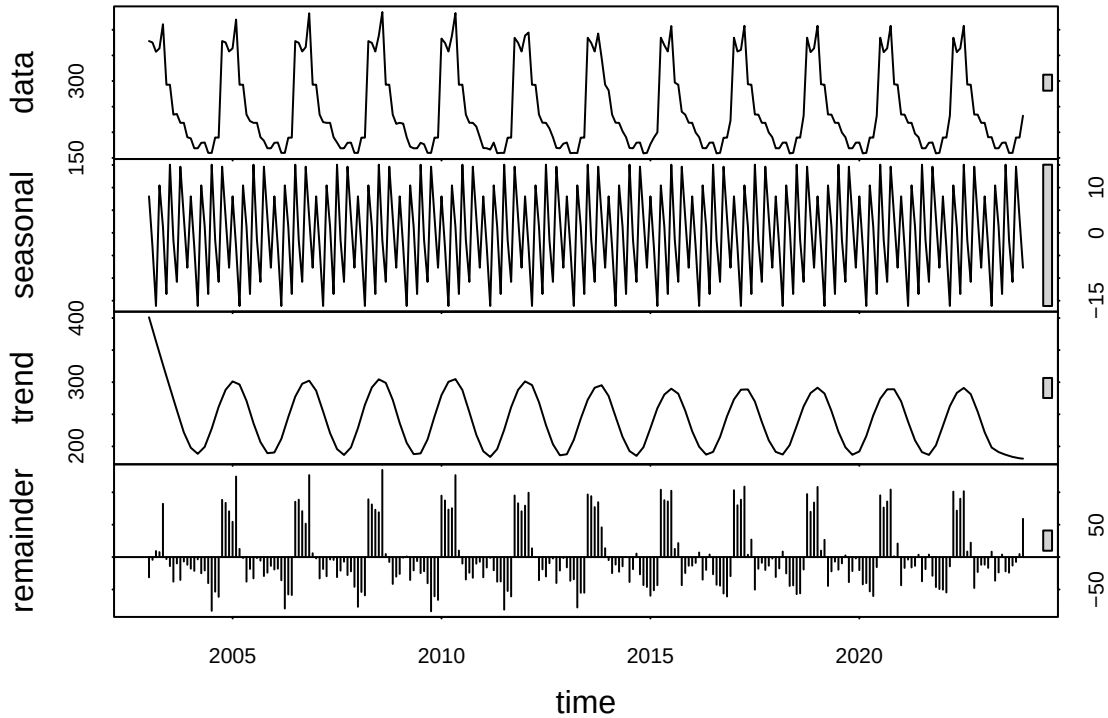

```
# Monthly Ozone for Delhi
Delhi.Ozone.monthly.ts <- ts(delhi_final_processed$O3,
                             start = c(2003,1), frequency = 12)
Delhi.Ozone.monthly.decomposed <- stl(Delhi.Ozone.monthly.ts,
                                      s.window = "periodic")
plot(Delhi.Ozone.monthly.decomposed)
```



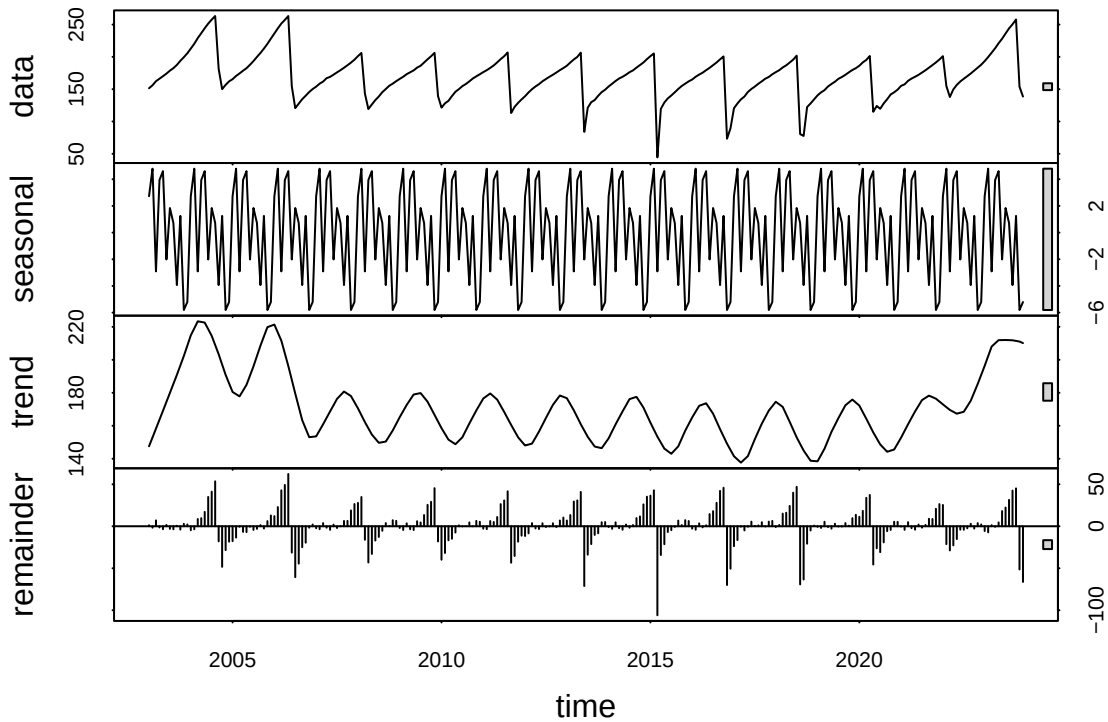

```r
# Monthly PM2.5 for Delhi
Delhi.PM2.5.monthly.ts <- ts(delhi_final_processed$PM2.5,
                             start = c(2003,1), frequency = 12)
Delhi.PM2.5.monthly.decomposed <- stl(Delhi.PM2.5.monthly.ts,
                                      s.window = "periodic")
plot(Delhi.PM2.5.monthly.decomposed)
```



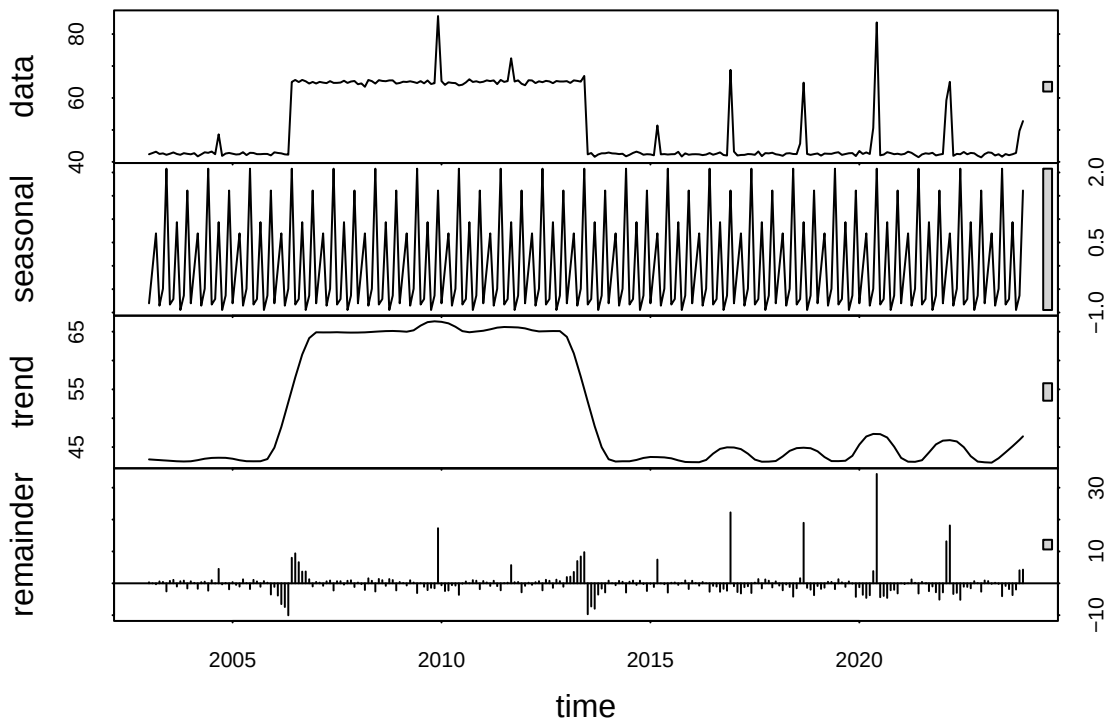

```r
# Monthly PM10 for Delhi
Delhi.PM10.monthly.ts <- ts(delhi_final_processed$PM10,
                            start = c(2003,1), frequency = 12)
Delhi.PM10.monthly.decomposed <- stl(Delhi.PM10.monthly.ts,
                                     s.window = "periodic")
plot(Delhi.PM10.monthly.decomposed)
```



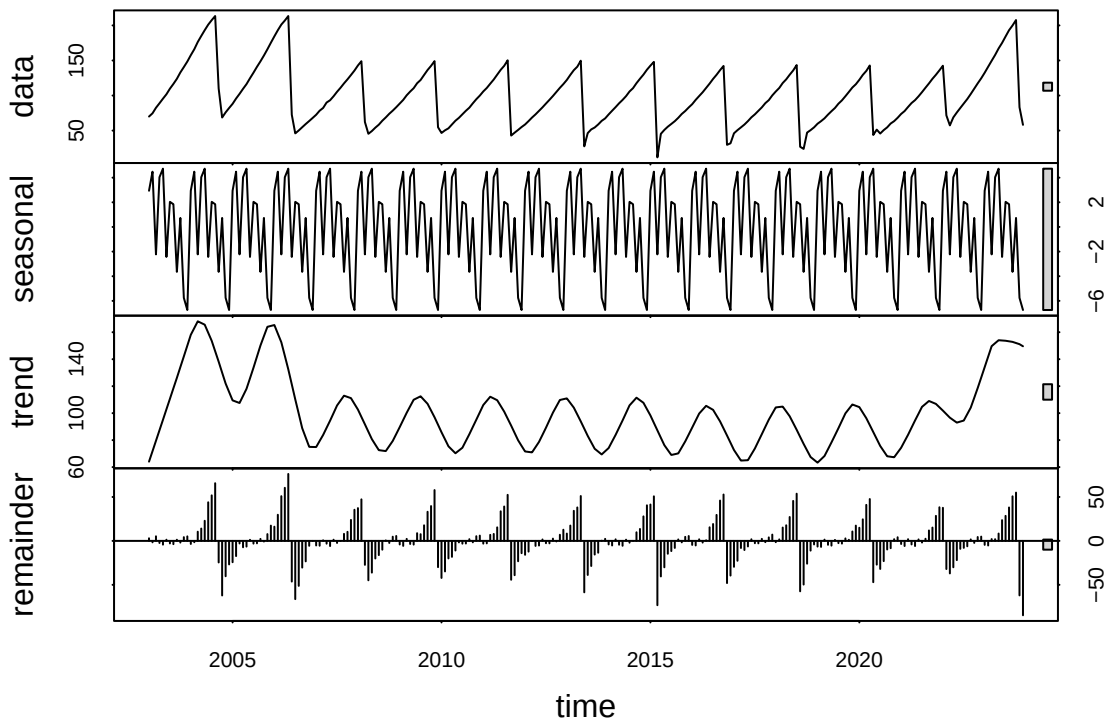

```
# Monthly AQI for Dhaka
Dhaka.AQI.monthly.ts <- ts(dhaka_final_processed$AQI,
                           start = c(2003,1), frequency = 12)
Dhaka.AQI.monthly.decomposed <- stl(Dhaka.AQI.monthly.ts,
                                    s.window = "periodic")
plot(Dhaka.AQI.monthly.decomposed)
```



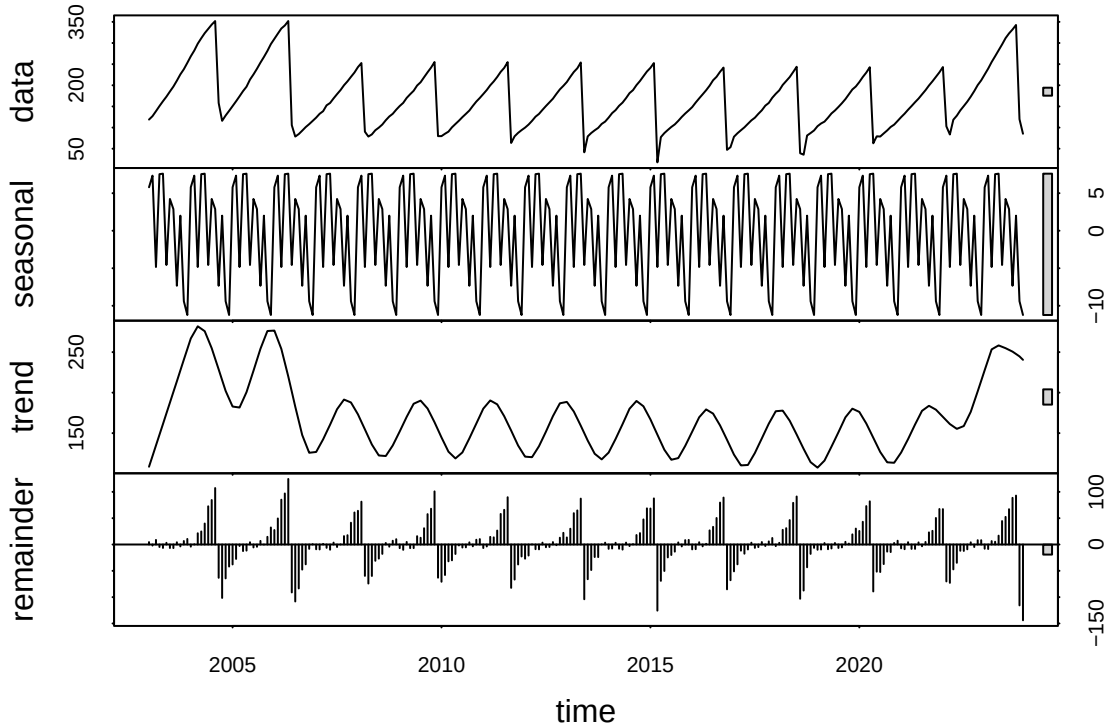

```r
# Monthly Ozone for Dhaka
Dhaka.Ozone.monthly.ts <- ts(dhaka_final_processed$O3,
                             start = c(2003,1), frequency = 12)
Dhaka.Ozone.monthly.decomposed <- stl(Dhaka.Ozone.monthly.ts,
                                      s.window = "periodic")
plot(Dhaka.Ozone.monthly.decomposed)
```




```r
# Monthly PM2.5 for Dhaka
Dhaka.PM2.5.monthly.ts <- ts(dhaka_final_processed$PM2.5,
                             start = c(2003,1), frequency = 12)
Dhaka.PM2.5.monthly.decomposed <- stl(Dhaka.PM2.5.monthly.ts,
                                      s.window = "periodic")
plot(Dhaka.PM2.5.monthly.decomposed)
```




```r
# Monthly PM10 for Dhaka
Dhaka.PM10.monthly.ts <- ts(dhaka_final_processed$PM10,
                            start = c(2003,1), frequency = 12)
Dhaka.PM10.monthly.decomposed <- stl(Dhaka.PM10.monthly.ts,
                                     s.window = "periodic")
plot(Dhaka.PM10.monthly.decomposed)
```




```r
# Seasonal Mann-Kendall Dhaka

# Dhaka AQI
Dhaka.AQI.monthly.trend <- trend::smk.test(Dhaka.AQI.monthly.ts)
Dhaka.AQI.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.AQI.monthly.ts
## z = -1.7696, p-value = 0.0768
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -204 13160
```

```r
summary(Dhaka.AQI.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.AQI.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
```

```
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -2 1096.7 -0.010 -0.030  0.97591
## Season 2:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 3:   S = 0  -14 1096.7 -0.067 -0.393  0.69464
## Season 4:   S = 0  -10 1096.7 -0.048 -0.272  0.78580
## Season 5:   S = 0  -34 1096.7 -0.162 -0.996  0.31901
## Season 6:   S = 0  -20 1096.7 -0.095 -0.574  0.56614
## Season 7:   S = 0   -6 1096.7 -0.029 -0.151  0.87999
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 10:   S = 0   6 1096.7  0.029  0.151  0.87999
## Season 11:   S = 0 -34 1096.7 -0.162 -0.996  0.31901
## Season 12:   S = 0 -30 1096.7 -0.143 -0.876  0.38119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka AQI not significant (p = 0.0768)


# Dhaka PM2.5
Dhaka.PM2.5.monthly.trend <- trend::smk.test(Dhaka.PM2.5.monthly.ts)
Dhaka.PM2.5.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.PM2.5.monthly.ts
## z = -1.7696, p-value = 0.0768
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -204 13160
```

```r
summary(Dhaka.PM2.5.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.PM2.5.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0    0 1096.7  0.000  0.000  1.00000
## Season 2:   S = 0  -28 1096.7 -0.133 -0.815  0.41489
## Season 3:   S = 0  -14 1096.7 -0.067 -0.393  0.69464
## Season 4:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 5:   S = 0  -38 1096.7 -0.181 -1.117  0.26387
## Season 6:   S = 0  -22 1096.7 -0.105 -0.634  0.52599
```

```
## Season 7:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 10:   S = 0   8 1096.7  0.038  0.211  0.83259
## Season 11:   S = 0 -28 1096.7 -0.133 -0.815  0.41489
## Season 12:   S = 0 -24 1096.7 -0.114 -0.695  0.48735
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka PM2.5 not significant (p = 0.0768)

# Dhaka PM10
Dhaka.PM10.monthly.trend <- trend::smk.test(Dhaka.PM10.monthly.ts)
Dhaka.PM10.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.PM10.monthly.ts
## z = -1.6127, p-value = 0.1068
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -186 13160
```

```r
summary(Dhaka.PM10.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.PM10.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0    4 1096.7  0.019  0.091  0.92782
## Season 2:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 3:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 4:   S = 0  -12 1096.7 -0.057 -0.332  0.73976
## Season 5:   S = 0  -32 1096.7 -0.152 -0.936  0.34922
## Season 6:   S = 0  -16 1096.7 -0.076 -0.453  0.65058
## Season 7:   S = 0   -8 1096.7 -0.038 -0.211  0.83259
## Season 8:   S = 0  -26 1096.7 -0.124 -0.755  0.45030
## Season 9:   S = 0   -2 1096.7 -0.010 -0.030  0.97591
## Season 10:   S = 0   6 1096.7  0.029  0.151  0.87999
## Season 11:   S = 0 -34 1096.7 -0.162 -0.996  0.31901
## Season 12:   S = 0 -28 1096.7 -0.133 -0.815  0.41489
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka PM2.5 not significant (p = 0.1068)

# Dhaka O3
Dhaka.Ozone.monthly.trend <- trend::smk.test(Dhaka.Ozone.monthly.ts)
Dhaka.Ozone.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Dhaka.Ozone.monthly.ts
## z = -4.8206, p-value = 1.432e-06
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##     S  varS
##  -554 13160
```

```r
summary(Dhaka.Ozone.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Dhaka.Ozone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0  -30 1096.7 -0.143 -0.876 0.381187
## Season 2:   S = 0  -38 1096.7 -0.181 -1.117 0.263872
## Season 3:   S = 0    6 1096.7  0.029  0.151 0.879988
## Season 4:   S = 0  -56 1096.7 -0.267 -1.661 0.096747 .
## Season 5:   S = 0  -54 1096.7 -0.257 -1.600 0.109502
## Season 6:   S = 0  -40 1096.7 -0.190 -1.178 0.238924
## Season 7:   S = 0  -70 1096.7 -0.333 -2.084 0.037198 *
## Season 8:   S = 0  -76 1096.7 -0.362 -2.265 0.023527 *
## Season 9:   S = 0  -62 1096.7 -0.295 -1.842 0.065473 .
## Season 10:   S = 0 -12 1096.7 -0.057 -0.332 0.739764
## Season 11:   S = 0 -66 1096.7 -0.314 -1.963 0.049669 *
## Season 12:   S = 0 -56 1096.7 -0.267 -1.661 0.096747 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Dhaka ozone (O3) is significant (1.432e-06)

# Seasonal Mann-Kendall Delhi

# Delhi AQI
Delhi.AQI.monthly.trend <- trend::smk.test(Delhi.AQI.monthly.ts)
Delhi.AQI.monthly.trend
```

```
##
```

```
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.AQI.monthly.ts
## z = -0.98905, p-value = 0.3226
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##  -114.00 13053.33
```

```r
summary(Delhi.AQI.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.AQI.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -8 1092.0 -0.038 -0.212  0.83224
## Season 2:   S = 0  -41 1088.3 -0.199 -1.212  0.22532
## Season 3:   S = 0    1 1085.7  0.005  0.000  1.00000
## Season 4:   S = 0   -6 1092.0 -0.029 -0.151  0.87973
## Season 5:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 6:   S = 0    9 1083.7  0.044  0.243  0.80799
## Season 7:   S = 0   16 1090.0  0.077  0.454  0.64959
## Season 8:   S = 0  -27 1088.3 -0.131 -0.788  0.43063
## Season 9:   S = 0    3 1083.7  0.015  0.061  0.95155
## Season 10:   S = 0  -4 1092.0 -0.019 -0.091  0.92766
## Season 11:   S = 0 -36 1084.7 -0.176 -1.063  0.28791
## Season 12:   S = 0  10 1084.7  0.049  0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi AQI not significant (0.3226)

# Delhi PM2.5
Delhi.PM2.5.monthly.trend <- trend::smk.test(Delhi.PM2.5.monthly.ts)
Delhi.PM2.5.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.PM2.5.monthly.ts
## z = -1.7156, p-value = 0.08624
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##  -197.00 13052.33
```

```r
summary(Delhi.PM2.5.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.PM2.5.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0  -24 1092.0 -0.115 -0.696  0.48642
## Season 2:   S = 0  -41 1088.3 -0.199 -1.212  0.22532
## Season 3:   S = 0  -13 1085.7 -0.063 -0.364  0.71571
## Season 4:   S = 0  -14 1092.0 -0.067 -0.393  0.69403
## Season 5:   S = 0  -19 1088.3 -0.092 -0.546  0.58533
## Season 6:   S = 0    3 1083.7  0.015  0.061  0.95155
## Season 7:   S = 0   -4 1090.0 -0.019 -0.091  0.92760
## Season 8:   S = 0  -13 1088.3 -0.063 -0.364  0.71605
## Season 9:   S = 0   -1 1083.7 -0.005  0.000  1.00000
## Season 10:   S = 0 -24 1092.0 -0.115 -0.696  0.48642
## Season 11:   S = 0 -37 1083.7 -0.181 -1.094  0.27413
## Season 12:   S = 0 -10 1084.7 -0.049 -0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi AQI not significant (0.08624)

# Delhi PM10
Delhi.PM10.monthly.trend <- trend::smk.test(Delhi.PM10.monthly.ts)
Delhi.PM10.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.PM10.monthly.ts
## z = -0.79649, p-value = 0.4257
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##   -92.00 13053.33
```

```r
summary(Delhi.PM10.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.PM10.monthly.ts
## alternative hypothesis: two.sided
##
```

```
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   -2 1092.0 -0.010 -0.030  0.97586
## Season 2:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 3:   S = 0   -5 1085.7 -0.024 -0.121  0.90338
## Season 4:   S = 0    2 1092.0  0.010  0.030  0.97586
## Season 5:   S = 0  -25 1088.3 -0.121 -0.727  0.46692
## Season 6:   S = 0    7 1083.7  0.034  0.182  0.85537
## Season 7:   S = 0   16 1090.0  0.077  0.454  0.64959
## Season 8:   S = 0  -31 1088.3 -0.150 -0.909  0.36316
## Season 9:   S = 0   -5 1083.7 -0.024 -0.122  0.90329
## Season 10:   S = 0   2 1092.0  0.010  0.030  0.97586
## Season 11:   S = 0 -30 1084.7 -0.146 -0.881  0.37857
## Season 12:   S = 0  10 1084.7  0.049  0.273  0.78464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi PM10 not significant (0.4257)

# Delhi O3
Delhi.Ozone.monthly.trend <- trend::smk.test(Delhi.Ozone.monthly.ts)
Delhi.Ozone.monthly.trend
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data:  Delhi.Ozone.monthly.ts
## z = 1.0157, p-value = 0.3098
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##        S     varS
##   117.00 13044.33
```

```r
summary(Delhi.Ozone.monthly.trend)
```

```
##
##  Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: Delhi.Ozone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##                      S   varS    tau      z Pr(>|z|)
## Season 1:   S = 0   25 1091.0  0.120  0.727  0.46747
## Season 2:   S = 0   11 1088.3  0.053  0.303  0.76180
## Season 3:   S = 0   14 1084.7  0.068  0.395  0.69305
## Season 4:   S = 0   19 1091.0  0.092  0.545  0.58578
## Season 5:   S = 0    9 1088.3  0.044  0.242  0.80839
## Season 6:   S = 0   14 1082.7  0.069  0.395  0.69278
```

```
## Season 7:   S = 0   21 1089.0  0.102  0.606  0.54447
## Season 8:   S = 0    8 1087.3  0.039  0.212  0.83189
## Season 9:   S = 0    9 1083.7  0.044  0.243  0.80799
## Season 10:   S = 0   5 1091.0  0.024  0.121  0.90361
## Season 11:   S = 0  -7 1083.7 -0.034 -0.182  0.85537
## Season 12:   S = 0 -11 1083.7 -0.054 -0.304  0.76130
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Delhi ozone (O3) not significant (0.3098)


# Non-Seasonal Mann-Kendall Dhaka

# Dhaka AQI
Dhaka.monthly.AQI.nonseasonal <- Dhaka.AQI.monthly.ts -
  Dhaka.AQI.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.AQI.trend <- Kendall::MannKendall(
  Dhaka.monthly.AQI.nonseasonal)
Dhaka.nonseasonal.AQI.trend
```

```
## tau = -0.0537, 2-sided pvalue =0.20448
```

```r
summary(Dhaka.nonseasonal.AQI.trend)
```

```
## Score =  -1698 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0537, 2-sided pvalue =0.20448
```

```r
# Dhaka PM2.5
Dhaka.monthly.PM2.5.nonseasonal <- Dhaka.PM2.5.monthly.ts -
  Dhaka.PM2.5.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.PM2.5.trend <- Kendall::MannKendall(
  Dhaka.monthly.PM2.5.nonseasonal)
Dhaka.nonseasonal.PM2.5.trend
```

```
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
summary(Dhaka.nonseasonal.PM2.5.trend)
```

```
## Score =  -1588 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
# Dhaka PM10
Dhaka.monthly.PM10.nonseasonal <- Dhaka.PM10.monthly.ts -
  Dhaka.PM10.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.PM10.trend <- Kendall::MannKendall(
  Dhaka.monthly.PM10.nonseasonal)
Dhaka.nonseasonal.PM10.trend
```

```
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
summary(Dhaka.nonseasonal.PM10.trend)
```

```
## Score =  -1588 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.0502, 2-sided pvalue =0.23537
```

```r
# Dhaka O3
Dhaka.monthly.Ozone.nonseasonal <- Dhaka.Ozone.monthly.ts -
  Dhaka.Ozone.monthly.decomposed$time.series[,1]

Dhaka.nonseasonal.Ozone.trend <- Kendall::MannKendall(
  Dhaka.monthly.Ozone.nonseasonal)
Dhaka.nonseasonal.Ozone.trend
```

```
## tau = -0.201, 2-sided pvalue =2.0616e-06
```

```r
summary(Dhaka.nonseasonal.Ozone.trend)
```

```
## Score =  -6350 , Var(Score) = 1788626
## denominator =  31626
## tau = -0.201, 2-sided pvalue =2.0616e-06
```

```r
# Non-Seasonal Mann-Kendall Delhi

# Delhi AQI
Delhi.monthly.AQI.nonseasonal <- Delhi.AQI.monthly.ts -
  Delhi.AQI.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.AQI.trend <- Kendall::MannKendall(
  Delhi.monthly.AQI.nonseasonal)
Delhi.nonseasonal.AQI.trend
```

```
## tau = -0.0681, 2-sided pvalue =0.10775
```

```r
summary(Delhi.nonseasonal.AQI.trend)
```

```
## Score =  -2152 , Var(Score) = 1788519
## denominator =  31580.97
## tau = -0.0681, 2-sided pvalue =0.10775
```

```r
# Delhi PM2.5
Delhi.monthly.PM2.5.nonseasonal <- Delhi.PM2.5.monthly.ts -
  Delhi.PM2.5.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.PM2.5.trend <- Kendall::MannKendall(
  Delhi.monthly.PM2.5.nonseasonal)
Delhi.nonseasonal.PM2.5.trend
```

```
## tau = -0.0903, 2-sided pvalue =0.03296
```

```r
summary(Delhi.nonseasonal.PM2.5.trend)
```

```
## Score =  -2853 , Var(Score) = 1788518
## denominator =  31580.47
## tau = -0.0903, 2-sided pvalue =0.03296
```

```r
# Delhi PM10
Delhi.monthly.PM10.nonseasonal <- Delhi.PM10.monthly.ts -
  Delhi.PM10.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.PM10.trend <- Kendall::MannKendall(
  Delhi.monthly.PM10.nonseasonal)
Delhi.nonseasonal.PM10.trend
```

```
## tau = -0.0624, 2-sided pvalue =0.14094
```

```r
summary(Delhi.nonseasonal.PM10.trend)
```

```
## Score =  -1970 , Var(Score) = 1788519
## denominator =  31580.97
## tau = -0.0624, 2-sided pvalue =0.14094
```

```r
# Delhi O3
Delhi.monthly.Ozone.nonseasonal <- Delhi.Ozone.monthly.ts -
  Delhi.Ozone.monthly.decomposed$time.series[,1]

Delhi.nonseasonal.Ozone.trend <- Kendall::MannKendall(
  Delhi.monthly.Ozone.nonseasonal)
Delhi.nonseasonal.Ozone.trend
```

```
## tau = 0.0326, 2-sided pvalue =0.44208
```

```r
summary(Delhi.nonseasonal.Ozone.trend)
```

```
## Score =  1029 , Var(Score) = 1788510
## denominator =  31576.46
## tau = 0.0326, 2-sided pvalue =0.44208
```

Table 1: Summary of Mann-Kendall Results

Location	Variable	Seasonal p-value	Non-Seasonal p-value	Trend Statistic
Dhaka	AQI	0.0768	0.2045	-204
Dhaka	PM 2.5	0.0768	0.2354	-204
Dhaka	PM 10	0.1068	0.2354	-186
Dhaka	O3	**1.43e-06**	**2.06e-06**	-554
Delhi	AQI	0.3226	0.1078	-114
Delhi	PM 2.5	0.0862	**0.0330**	-197

Location	Variable	Seasonal p-value	Non-Seasonal p-value	Trend Statistic
Delhi	PM 10	0.4257	0.1409	-92
Delhi	O3	0.3098	0.4421	117

```r
# AQI
t.test(AQI ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  AQI by Location
## t = 19.658, df = 344.72, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##   94.82248 115.90689
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            275.7138            170.3491
```

```r
# PM2.5
t.test(PM2.5 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  PM2.5 by Location
## t = 35.011, df = 490.03, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  132.4543 148.2050
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            240.5413            100.2117
```

```r
# PM10
t.test(PM10 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  PM10 by Location
## t = 10.773, df = 490.34, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  59.67191 86.29319
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            242.1669            169.1844
```

```r
# O3
t.test(O3 ~ Location, data = combined_dataset)
```

```
##
##  Welch Two Sample t-test
##
## data:  O3 by Location
## t = -15.654, df = 450.94, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Delhi and group Dhaka is not equal to
## 95 percent confidence interval:
##  -21.48934 -16.69565
## sample estimates:
## mean in group Delhi mean in group Dhaka
##            31.78149            50.87398
```

Table 2: Summary of two sample t-test Results

Variable	p-value	Dhaka mean	Delhi mean
AQI	**2.2e-16**	170.35	275.71
PM2.5	**2.2e-16**	100.21	240.54
PM10	**2.2e-16**	169.18	242.17
O3	**2.2e-16**	50.87	31.78

```r
#Linear Regression

# AQI
AQI.model <- lm(AQI ~ first.month * Location, data = combined_dataset)
summary(AQI.model)
```

```
##
## Call:
## lm(formula = AQI ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -152.894  -19.418   -1.296   16.357   93.331
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.595e+02  1.582e+01   48.00   <2e-16 ***
## first.month               -3.048e-02  9.873e-04  -30.87   <2e-16 ***
## LocationDhaka             -7.062e+02  2.238e+01  -31.56   <2e-16 ***
## first.month:LocationDhaka  3.785e-02  1.396e-03   27.11   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.7 on 500 degrees of freedom
## Multiple R-squared:  0.8128, Adjusted R-squared:  0.8117
## F-statistic: 723.5 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# PM2.5
PM2.5.model <- lm(PM2.5 ~ first.month * Location, data = combined_dataset)
summary(PM2.5.model)
```

```
##
## Call:
## lm(formula = PM2.5 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -127.439  -13.193   -3.511    9.854   79.986
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               5.623e+02  1.238e+01   45.42   <2e-16 ***
## first.month              -2.027e-02  7.725e-04  -26.24   <2e-16 ***
## LocationDhaka            -6.310e+02  1.751e+01  -36.04   <2e-16 ***
## first.month:LocationDhaka  3.091e-02  1.092e-03   28.30   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.15 on 500 degrees of freedom
## Multiple R-squared:  0.8946, Adjusted R-squared:  0.894
## F-statistic:  1415 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# PM10
PM10.model <- lm(PM10 ~ first.month * Location, data = combined_dataset)
summary(PM10.model)
```

```
##
## Call:
## lm(formula = PM10 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -214.902  -26.310   -5.708   23.492  128.747
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               7.438e+02  2.324e+01   32.00   <2e-16 ***
## first.month              -3.160e-02  1.450e-03  -21.79   <2e-16 ***
## LocationDhaka            -8.491e+02  3.287e+01  -25.83   <2e-16 ***
## first.month:LocationDhaka  4.890e-02  2.051e-03   23.84   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50.98 on 500 degrees of freedom
## Multiple R-squared:  0.6364, Adjusted R-squared:  0.6342
## F-statistic: 291.7 on 3 and 500 DF,  p-value: < 2.2e-16
```

```r
# O3
O3.model <- lm(O3 ~ first.month * Location, data = combined_dataset)
summary(O3.model)
```

```
##
## Call:
## lm(formula = O3 ~ first.month * Location, data = combined_dataset)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -30.486  -9.189  -4.640  13.627  32.740
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -2.075e+01  5.773e+00  -3.595 0.000356 ***
## first.month                3.310e-03  3.602e-04   9.189  < 2e-16 ***
## LocationDhaka              6.263e+01  8.164e+00   7.671 8.97e-14 ***
## first.month:LocationDhaka -2.743e-03  5.094e-04  -5.384 1.12e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.66 on 500 degrees of freedom
## Multiple R-squared:  0.4275, Adjusted R-squared:  0.4241
## F-statistic: 124.5 on 3 and 500 DF,  p-value: < 2.2e-16
```

Table 3: Summary of Linear Regression Results

Variable	p-value	Min	Median	Max	R-squared
AQI	**2.2e-16**	-152.894	-1.296	93.331	0.812
PM2.5	**2.2e-16**	-127.439	-3.511	79.986	0.894
PM10	**2.2e-16**	-214.902	-5.708	128.747	0.634
O3	**2.2e-16**	-30.487	-4.640	32.740	0.424